Page a day worksheet **PAD nr: 1**

Grade 10 – Graphs and Equations – 1

Name:	

QUESTION 1

Solve for x in the following equations. Show ALL calculations.

(a) $x^2 - 3x = 4$ (2)

(b) $3^{x-1} - 81 = 0$ (3)

[5]

QUESTION 2

Given: $f(x) = 3x^2 - 27$

(a) Calculate the intercepts with the axes. (3)

(b) Give the range of the graph. (2)

(c) Make a neat sketch of the graph below. (4)

(d) For which values of x will the graph be decreasing? (2)

[11]

Page a day worksheet **PAD nr: 2**

Grade 10 – Graphs and Equations – 2

Name:	

QUESTION 1

Solve for x in the following equations. Show ALL calculations.

(a) $\left(\frac{1}{2}\right)^x = 64$ (2)

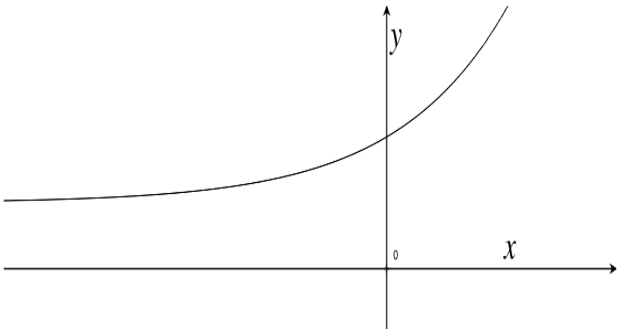
(c) $(x-2)(x+3) = 6$ (3)

[5]

QUESTION 2

The graph of $f(x) = a^x + q$ is drawn:

The graph has an asymptote at $y = 1$ and there is a set of coordinates (2;5) on the graph. Answer the questions below:



(a) Give the range of the graph. (2)

(b) Determine the values of a and q. (3)

(c) Give the equation reflection of the graph in the x-axis (2)

(d) How must the graph be translated to be $y = 2^x - 3$? (2)

[11]

Page a day worksheet **PAD nr: 3**

Grade 10 – Graphs and Equations – 3

Name:	

QUESTION 1

Solve for x in the following equations. Show ALL calculations.

(a) $\frac{3}{x} = \frac{x-1}{2}$ (4)

(c) $25^x - \frac{1}{125} = 0$ (2)

[6]

QUESTION 2

You are given the function of $f(x) = \frac{-3}{x} + 2$. Answer the following questions:

(a) Give the equations of the asymptotes of f. (2)

(b) Determine the x-intercept of f. (2)

(c) If the coordinates $(-1; a)$, $(-2; b)$ and $(3; c)$ lie on f, determine the values of a, b and c. (3)

(d) Draw a sketch of f and clearly indicate all intercepts with the axes, asymptotes and also any other relevant points. (4)

[12]